

SAMAY THAKUR

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EDUCATION

M.S. Ramaiah Institute of Technology

Bachelor of Engineering | Electronics and Instrumentation | CGPA: 7.95

Oct 2023 – Present

Bengaluru, India

Sunbeam School Varuna

Grade X (CBSE): 89.6% | Grade XII (CBSE): 74.2%

Mar 2019 – Jun 2022

Varanasi, India

EXPERIENCE

CPS & IoT Intern

Aug 2024 – Aug 2024

Intra Institutional Internship

Bengaluru, India

- Developed a real-time water-level monitoring and automated pump-control system using Arduino Uno and ultrasonic sensors, implementing sensor data acquisition and threshold-based control logic in C++.
- Tested and debugged the system across varying water-level conditions to automate pump ON/OFF operation and prevent tank overflow and dry-run conditions.

PROJECTS

SmartQC: On-Device Visual Quality Inspection | GitHub

Qualcomm | Snapdragon® AI Lab Build & Present Challenge

- Defined product scope and target user (small food producers needing visual QC without industrial hardware costs); built an on-device inspection system trained to 99.2% validation accuracy (0.9997 ROC-AUC) using Qualcomm AI Hub for Snapdragon.
- Evaluated the model beyond accuracy (confusion matrix, precision/recall, ROC-AUC) and stress-tested against 352 uncured real-world images, achieving 97.4% accuracy to inform a confidence-threshold feature for flagging low-confidence predictions.
- Prioritized and shipped features (explainability, batch processing, confidence flagging) based on real workflow needs, documenting the build, trade-offs, and limitations in a technical report for reproducibility.
- Technologies/Tools: Python, PyTorch, ONNX, Qualcomm AI Hub, PyQt5, OpenCV, scikit-learn

CynLr EPM PoC | GitHub

- Developed an end-to-end Engineering Program Management plan for a robotics PoC spanning moving-object tracking, track-and-grasp, hardware integration, testing, and delivery.
- Structured 41 top-level activities and 97 subtasks in Asana, mapping dependencies, milestones, risks, constraints, and external dependencies to establish an execution framework.
- Mapped workstreams across robotics, computer vision/ML, software, mechanical, electronics, procurement, and external vendors, developing a preliminary 604 person-hour effort model for resource planning.
- Designed a KPI and progress-monitoring framework covering dependency resolution, development progress, infrastructure readiness, testing outcomes, and delivery readiness.
- Technologies/Tools: Asana, Microsoft Excel

AI-Based Bread Quality Indicator System | GitHub

- Developed an end-to-end bread-quality assessment system combining MQ-135 gas sensing and ESP32-CAM image capture for real-time quality evaluation.
- Built a MobileNetV2-based image classification pipeline with 128×128 preprocessing, achieving 100% validation accuracy after 15 training epochs.
- Integrated model predictions, Wi-Fi telemetry, and automated LED/buzzer alerts through a Flask-based monitoring interface to provide real-time quality feedback.
- Technologies/Tools: Python, TensorFlow, MobileNetV2, TensorFlow Lite, Flask, ESP32-CAM

Stock Market Price Prediction & Analysis | GitHub

- Developed an LSTM-based forecasting model to predict next-day NIFTY 50 price movements using historical market data.
- Engineered predictive features using RSI, MACD, Moving Averages, and Bollinger Bands to capture market trends and support next-day forecasting.
- Achieved 1.39% MAPE and 93%+ R^2 on next-day price prediction, evaluating performance against baseline forecasting models.
- Technologies/Tools: Python, TensorFlow, LSTM, Pandas, NumPy, Scikit-learn

TECHNICAL SKILLS

Programming Languages: Python, C++

Data & Analytics: SQL (Basic), Microsoft Excel, Tableau, Power BI (Basic)

AI/ML: PyTorch, TensorFlow, TensorFlow Lite, Scikit-learn, Computer Vision, Deep Learning

Product/Program Management: Product Scoping, Feature Prioritization, Project Planning, Dependency Management, Risk Management, Resource Planning, KPI Definition, Cross-functional Coordination

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Flask, ONNX, Qualcomm AI Hub, OpenCV, PyQt5, SQLite

Concepts: Algorithms, Data Structures, Compiler Design, Computer Networks, Database Management Systems, Digital Logic Design, Operating System

ACHIEVEMENTS & LEADERSHIP

- Student Member:** Instrument Society of India (ISOI), IISc Bangalore (Nov 2023 – Oct 2027)
- Community Volunteer:** Organized and supported blood-donation drives through school and community initiatives, facilitating donor engagement and community welfare.